

1

Problem

ENCODE A DISTRIBUTION

- Directed models with continuous latent variables often have an intractable marginal likelihood and posterior.
- Per-example MCMC or iterative variational updates are too expensive for large datasets and minibatch learning.

2

Probabilistic autoencoder

ENCODE A DISTRIBUTION

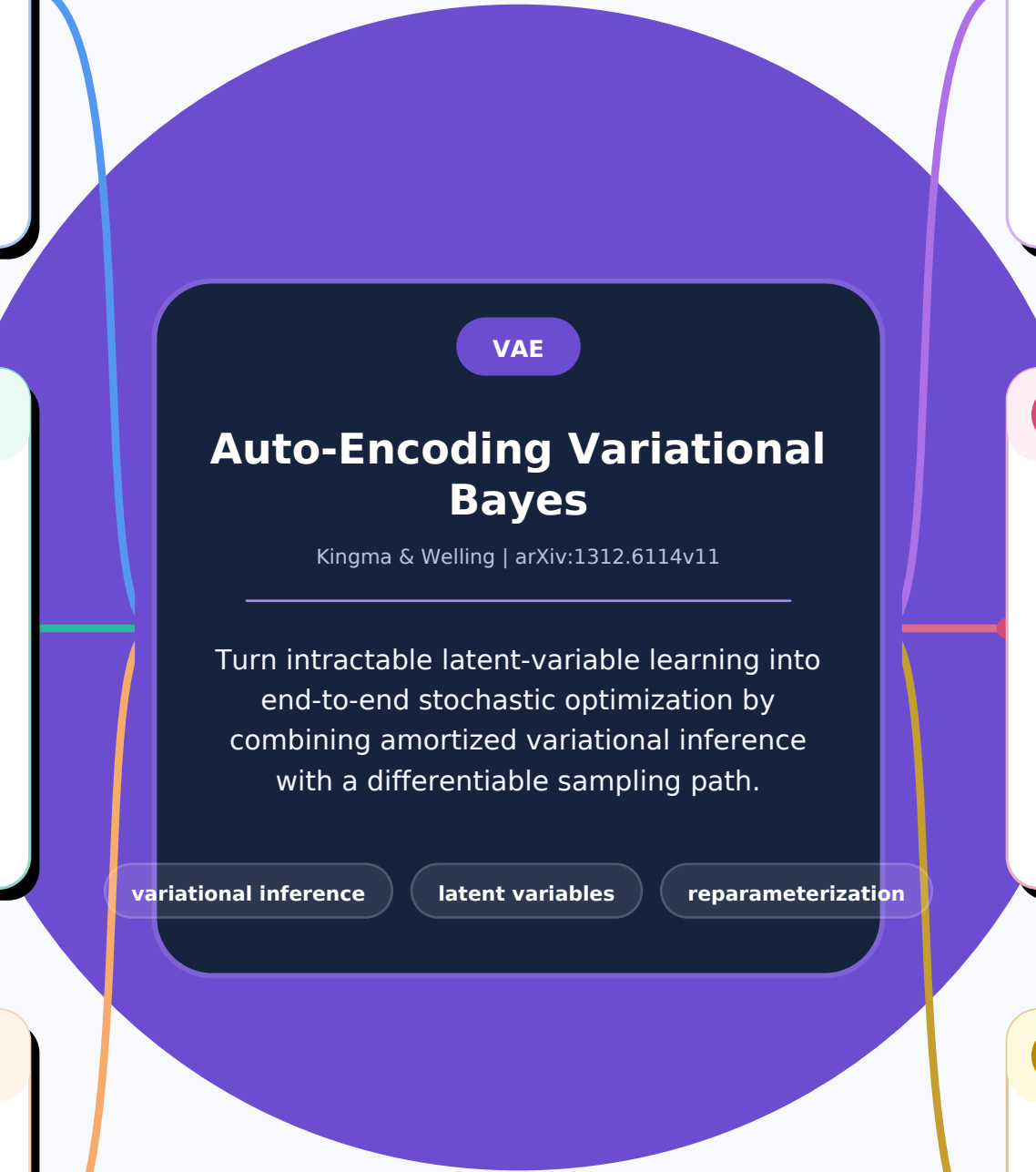
- The recognition model $q_{\phi}(z|x)$ amortizes posterior inference; the decoder $p_{\theta}(x|z)$ defines the generative likelihood.
- A standard Gaussian prior organizes a continuous latent code that supports sampling, denoising and visualization.

3

Learning objective

MAXIMIZE THE ELBO

- $ELBO = E_q[\log p_{\theta}(x|z)] - KL(q_{\phi}(z|x) || p(z))$
- The reconstruction term explains data; the KL term regularizes each approximate posterior toward the prior.



4

Key trick

MOVE RANDOMNESS OUTSIDE

- $z = \mu_{\phi}(x) + \sigma_{\phi}(x) * \epsilon; \epsilon \sim N(0, I)$
- Reparameterization produces a low-variance differentiable Monte Carlo estimator optimized with ordinary stochastic gradients.

5

Evidence

MNIST + FREY FACES

- AEVB converges faster and reaches a better variational bound than wake-sleep across tested latent dimensions.
- Experiments learn smooth 2-D manifolds, generate samples and scale with minibatches using about one sample per datapoint.

6

Meaning & limits

- The paper connects probabilistic inference and neural autoencoders, establishing the modern VAE recipe.
- The original estimator targets reparameterizable continuous latents; simple Gaussian decoders can yield blurry samples and loose bounds.